

Alexander Vega-Hernandez

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EDUCATION

University of Michigan–Ann Arbor

Bachelor of Science in Computer Science

Ann Arbor, MI

August 2025

Relevant Coursework: Distributed Systems, Algorithms and Data Structures, Databases, Web Systems, Computer Security, Computer Architecture, Software Engineering

TECHNICAL SKILLS

Languages: C++, C, Python, Go, JavaScript, SQL, HTML/CSS

Frameworks/Tools: React, Flask, Node.js, Docker, Git, Bash | **Databases:** PostgreSQL, SQLite, MySQL, MongoDB | **Cloud:** AWS

EXPERIENCE

R&D Software Engineer Intern

May 2026 – Present

Dassault Systèmes | C++, 3DExperience, XML

Royal Oak, MI

- Built production debug tooling for a multi-robot path planning system, including automated voxel pose updates and a route feasibility validator, ramping up on robotics and path-planning fundamentals independently in the process.
- Replaced a BFS-based pathfinding approach with a self-implemented Dijkstra's algorithm, now adopted as the default pathfinding method across the system; built real-time UI tooling for path inspection.
- Built a spatial awareness system computing a 3D distance transform and applying the Hamilton-Jacobi PDE to extract the Medial Axis, unblocking the team's Fast Marching Tree (FMT) planner integration and contributing to the project shipping roughly a month ahead of schedule.

PROJECTS

Sharded Key-Value Store with Paxos Consensus | Go, Distributed Systems

- Built a sharded, Paxos-based key-value store with leader election, replicated state management, and fault-tolerant request handling; debugged and revised replication logic after discovering consistency issues during testing.

Distributed MapReduce Framework | Python, Distributed Systems

- Implemented a Manager/Worker MapReduce runtime with TCP/UDP messaging for worker coordination and automatic task reassignment; revised partitioning logic after peer code review to improve reliability

Search Engine | Python, Flask, SQL, Web Systems

- Built a search engine over a Wikipedia dataset using a segmented inverted index with tf-idf scoring and PageRank-influenced ranking; developed Flask REST services and collaborated through GitHub to optimize query performance

LC2K Processor | C, Computer Architecture

- Designed and implemented both single-cycle and pipelined simulators for the LC2K instruction set architecture, including forwarding to handle pipeline hazards.

LEADERSHIP

Project Manager & Team Captain

June 2021 – August 2021

Henry Ford Learning Institute Summer Capstone

Detroit, MI

- Managed a 5-person team to design a college guidance app for Latinx students, refining layout based on user research, collaborated with cross-functional groups for design peer reviews, and pitched final blueprints to HFLI community leaders.

ADDITIONAL INFORMATION

Languages: English, Spanish (Bilingual)

Interests: R&D, Software Maintenance and Scalability, Open Source Contribution, UI/UX Design, CAD